

INPUTS FOR EQUIVARIANT INFINITE LOOP SPACE MACHINE

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ABSTRACT. Symmetric monoidal categories are central studies in modern algebraic topology, for instance K -theory relies on constructing spectra with suitable homotopy groups out of symmetric monoidal categories using the infinite loop space machine. It's known that the two inputs - pseudoalgebra over an operad $\mathcal{P}$, or suitable categories described by based finite sets $\mathcal{F}$ - are equivalent description for symmetric monoidal categories.

Within equivariant K -theory, similar machineries have been established, where the input for constructing G -spectra are described by pseudoalgebra over an operad $\mathcal{P}_G$ as G -categories, or categories described by based finite G -sets $\mathcal{F}_G$. The equivalence of the two within equivariant case is a natural question to ask, and associates to the question of “what is symmetric monoidal G -categories”.

We'll provide an exposition on the two inputs by collecting the axioms scattered (or implicitly assumed) in various literatures, while describe some equivariance properties of the two inputs.

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INTRODUCTION (ND)

1. CATEGORICAL PRELIMINARIES

1.1. 2-Categories.

Definition 1.1. A category $\mathcal{C}$ is a *2-category*, if for each object $X, Y \in \mathcal{C}$ (called *0-cells*), the homset $\mathcal{C}(X, Y)$ is itself a category (called the *hom-category*). Also, for every 0-cells $X, Y, Z \in \mathcal{C}$, the *composition* $\circ : \mathcal{C}(Y, Z) \times \mathcal{C}(X, Y) \rightarrow \mathcal{C}(X, Z)$ is a functor.

Every morphism $f \in \mathcal{C}(X, Y)$ is called a *1-cell*, while given any 1-cells $f, g \in \mathcal{C}(X, Y)$, a morphism $\alpha : f \Rightarrow g$ in the hom-category $\mathcal{C}(X, Y)$ is called a *2-cell*.

Here, $\circ$ must be strictly unital and associative.

Remark 1.2. There is a more general notion of a *bicategory*, which relaxes the composition functors to be associative and unital up to coherent natural isomorphisms, and in some literature this is called 2-categories. Here we'll take the strict associativity and unitality as axioms for 2-categories. For more general bicategory, cf. [7] Section 2.1.

Example 1.3. Let $\mathbf{Cat}$ denote the *category of small categories*, where the objects are small categories, the hom-categories $\mathbf{Cat}(\mathcal{A}, \mathcal{B})$ are functors between small categories $\mathcal{A}$ and $\mathcal{B}$, and the morphisms in $\mathbf{Cat}(\mathcal{A}, \mathcal{B})$ are natural transformations between functors. This specifies $\mathbf{Cat}$ to be a 2-category.

1.2. Pseudofunctor, Pseudotransform, and Modification.

Definition 1.4 (Pseudofunctor). Given two 2-categories $\mathcal{A}, \mathcal{B}$, a *pseudofunctor* $F : \mathcal{A} \rightarrow \mathcal{B}$ equips with an 0-cell map, functors between hom-categories $F_{X,Y} : \mathcal{A}(X, Y) \rightarrow \mathcal{B}(F(X), F(Y))$, an invertible 2-cell $\phi_{X,Y,Z}$ between compositions

$$\begin{array}{ccc} \mathcal{A}(Y, Z) \times \mathcal{A}(X, Y) & \xrightarrow{F_{Y,Z} \times F_{X,Y}} & \mathcal{B}(F(Y), F(Z)) \times \mathcal{B}(F(X), F(Y)) \\ \downarrow \circ & \nearrow \phi_{X,Y,Z} & \downarrow \circ \\ \mathcal{A}(X, Z) & \xrightarrow{F_{X,Z}} & \mathcal{B}(F(X), F(Z)) \end{array}$$

and another invertible 2-cell ψ_X between identities

$$\begin{array}{ccc} & \text{id}_{F(X)} & \\ & \downarrow & \\ F(X) & \xrightarrow{\psi_X} & F(X) \\ & \uparrow & \\ & F_{X,X}(\text{id}_X) & \end{array}$$

that satisfies:

(i) **(Associativity)** The following two pasting diagrams are equal:

$$\begin{array}{ccc}
 \mathcal{A}(Z, W) \times \mathcal{A}(Y, Z) \times \mathcal{A}(X, Y) & \xrightarrow{\Pi F_{--}} & \mathcal{B}(F(Z), F(W)) \times \mathcal{B}(F(Y), F(Z)) \times \mathcal{B}(F(X), F(Y)) \\
 \downarrow \circ \times \text{id} & \swarrow \phi_{Y, Z, W} \times \text{id} & \downarrow \circ \times \text{id} \\
 \mathcal{A}(Y, W) \times \mathcal{A}(X, Y) & \xleftarrow{F_{Y, W} \times F_{X, Y}} \mathcal{B}(F(Y), F(W)) \times \mathcal{B}(F(X), F(Y)) & \\
 \downarrow \circ & \swarrow \phi_{X, Y, W} & \downarrow \circ \\
 \mathcal{A}(X, W) & \xleftarrow{F_{X, W}} \mathcal{B}(F(X), F(W)) &
 \end{array}$$

equals

$$\begin{array}{ccc}
 \mathcal{A}(Z, W) \times \mathcal{A}(Y, Z) \times \mathcal{A}(X, Y) & \xrightarrow{\Pi F_{--}} & \mathcal{B}(F(Z), F(W)) \times \mathcal{B}(F(Y), F(Z)) \times \mathcal{B}(F(X), F(Y)) \\
 \downarrow \text{id} \times \circ & \swarrow \text{id} \times \phi_{X, Y, Z} & \downarrow \text{id} \times \circ \\
 \mathcal{A}(Z, W) \times \mathcal{A}(X, Z) & \xleftarrow{F_{Z, W} \times F_{X, Z}} \mathcal{B}(F(Z), F(W)) \times \mathcal{B}(F(X), F(Z)) & \\
 \downarrow \circ & \swarrow \phi_{X, Z, W} & \downarrow \circ \\
 \mathcal{A}(X, W) & \xleftarrow{F_{X, W}} \mathcal{B}(F(X), F(W)) &
 \end{array}$$

(ii) **(Unitality)** The following two unit diagrams commute:

$$\begin{array}{ccc}
 \mathcal{A}(X, Y) \times \mathcal{A}(X, X) & \xrightarrow{F_{X, Y} \times F_{X, X}} & \mathcal{B}(F(X), F(Y)) \times \mathcal{B}(F(X), F(X)) \\
 \uparrow \text{id} \times \text{id}_X & \swarrow \text{id} * \psi_X & \downarrow \circ \\
 \mathcal{A}(X, Y) \times * & & \\
 \downarrow F_{X, Y} \times \text{id}_{F(X)} & \swarrow & \downarrow \\
 \mathcal{B}(F(X), F(Y)) \times \mathcal{B}(F(X), F(X)) & \xrightarrow{\circ} & \mathcal{B}(F(X), F(Y))
 \end{array}$$

$$\begin{array}{ccc}
 \mathcal{A}(Y, Y) \times \mathcal{A}(X, Y) & \xrightarrow{F_{Y, Y} \times F_{X, Y}} & \mathcal{B}(F(Y), F(Y)) \times \mathcal{B}(F(X), F(Y)) \\
 \uparrow \text{id}_Y \times \text{id} & \swarrow \psi_Y * \text{id} & \downarrow \circ \\
 * \times \mathcal{A}(X, Y) & & \\
 \downarrow \text{id}_{F(Y)} \times F_{X, Y} & \swarrow & \downarrow \\
 \mathcal{B}(F(Y), F(Y)) \times \mathcal{B}(F(X), F(Y)) & \xrightarrow{\circ} & \mathcal{B}(F(X), F(Y))
 \end{array}$$

where $\text{id}_X : * \rightarrow \mathcal{A}(X, X)$ is the functor indicating identity of X .

If ψ_X is identity 2-cell for all 0-cell X , then the pseudofunctor F is called *normal*.

Definition 1.5 (Pseudotransform). Fix 2-categories $\mathcal{A}, \mathcal{B}$, let $(F, \phi, \psi), (G, \varphi, \vartheta) : \mathcal{A} \rightarrow \mathcal{B}$ be two pseudofunctors. A *pseudotransform* $\mu : (F, \phi, \psi) \rightarrow (G, \varphi, \vartheta)$ collects 1-cells $\mu_X : F(X) \rightarrow G(X)$ for each 0-cell $X \in \mathcal{A}$, and for each 1-cell

$f : X \rightarrow Y$ in $\mathcal{A}$, associates a 2-cell μf such that

$$\begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ Ff \downarrow & \mu f \swarrow & \searrow Gf \\ F(Y) & \xrightarrow{\mu_Y} & G(Y) \end{array}$$

commutes. They also satisfy:

- (i) (**Associativity**) Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be 1-cells in $\mathcal{A}$, the following two pasting diagrams are equal:

$$\begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow Ff & \searrow \mu f & \searrow Gf \\ F(Y) & \xrightarrow{\mu_Y} & G(Y) \\ \downarrow F(g \circ f) & \swarrow \phi_{g,f} & \swarrow \mu(g \circ f) \\ F(Z) & \xrightarrow{\mu_Z} & G(Z) \end{array} = \begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow F(g \circ f) & \swarrow \mu(g \circ f) & \swarrow G(g \circ f) \\ F(Z) & \xrightarrow{\mu_Z} & G(Z) \end{array}$$

- (ii) (**Unitality**) The following pasting diagrams are equal:

$$\begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow F(\text{id}_X) & \searrow \psi_X & \searrow \text{id}_{F(X)} \\ F(X) & \xrightarrow{\mu_X} & G(X) \end{array} = \begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow F(\text{id}_X) & \swarrow \mu(\text{id}_X) & \swarrow \text{id}_{G(X)} \\ F(X) & \xrightarrow{\mu_X} & G(X) \end{array}$$

Definition 1.6 (Modification). Continue with the same notation as above. Let $\mu, \nu : (F, \phi, \psi) \rightarrow (G, \varphi, \vartheta)$ be two pseudotransforms, a *modification* $\lambda : \mu \Rightarrow \nu$ is a collection of 2-cells $\lambda_X : \mu_X \Rightarrow \nu_X$ for each 0-cell $X \in \mathcal{A}$, such that the following two pasting diagrams are equal, for each 1-cell $f : X \rightarrow Y$ in $\mathcal{A}$:

$$\begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow Ff & \searrow \mu f & \searrow Gf \\ F(Y) & \xrightarrow{\mu_Y} & G(Y) \end{array} = \begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow Ff & \swarrow \mu f & \swarrow Gf \\ F(Y) & \xrightarrow{\mu_Y} & G(Y) \end{array}$$

1.3. G -Categories.

Definition 1.7. A small category $\mathcal{A}$ is a G -category, if the object and morphism sets $\text{Ob}(\mathcal{A}), \text{Mor}(\mathcal{A})$ are both (left) G -sets, such that the following four structure maps (S, T, I, C) (stands for *source, target, identity, and composition*) are all G -equivariant:

- $S, T : \text{Mor}(\mathcal{A}) \rightarrow \text{Ob}(\mathcal{A})$ by $S(X \xrightarrow{f} Y) := X$, and $T(X \xrightarrow{f} Y) := Y$.
- $I : \text{Ob}(\mathcal{A}) \rightarrow \text{Mor}(\mathcal{A})$ by $I(X) := \text{id}_X$.
- $C : \text{Mor}(\mathcal{A}) \times_{S, T} \text{Mor}(\mathcal{A}) \rightarrow \text{Mor}(\mathcal{A})$ by $C(g, f) := g \circ f$.

Here, $\text{Mor}(\mathcal{A}) \times_{S, T} \text{Mor}(\mathcal{A}) := \{(g, f) \in \text{Mor}(\mathcal{A})^2 \mid S(g) = T(f)\}$, precisely the pair of morphisms that're composable, and it endows G -set structure via restricting the diagonal G -action on $\text{Mor}(\mathcal{A})^2$. This is well-defined due to the assumption S, T are G -maps.

Another equivalent characterization can be viewed as a group homomorphism $G \rightarrow \text{Aut}_{\mathbf{Cat}}(\mathcal{A})$, which assigns each $g \in G$ an automorphism functor on $\mathcal{A}$ such that composition respects G 's multiplication structure, and identity gets sent to the identity functor. So, from now on we view each element $g \in G$ as a functor, whenever a category is specified as a G -category.

Definition 1.8. Let $\mathbf{Cat}_G$ denotes the 2-category of small G -categories, with (nonequivariant) functors as 1-cells, natural transformations as 2-cells. The hom-category between any two G -categories $\mathbf{Cat}_G(\mathcal{A}, \mathcal{B})$ also has a natural G -action via conjugation. Which, the G -fixed subcategory of the hom-category $[\mathbf{Cat}_G(\mathcal{A}, \mathcal{B})]^G$ precisely describes all G -functor, or the functors $F : \mathcal{A} \rightarrow \mathcal{B}$ such that $g \circ F = F \circ g$ for all $g \in G$.

Another important notion is the twisted G -action on products when given a group homomorphism $\alpha : G \rightarrow \Sigma_n$. Given a category $\mathcal{A}$, the n -fold product $\mathcal{A}^n$ has a natural left Σ_n -action via permuting the coordinates; if $\mathcal{A}$ is a G -category, $\mathcal{A}^n$ also endows a natural diagonal G -action. The two coordinates to give a natural left $(G \times \Sigma_n)$ -action on $\mathcal{A}^n$.

Definition 1.9. Let $\mathcal{A}^{n^\alpha}$ be a G -category with the underlying category being $\mathcal{A}^n$, with the G -action defined as follow:

$$g \cdot_\alpha (X_1, \dots, X_n) := (g \cdot X_{\alpha(g)^{-1}(1)}, \dots, g \cdot X_{\alpha(g)^{-1}(n)})$$

Another characterization can be understood as follow: Given α 's graph subgroup $\Gamma_\alpha := \{(g, \alpha(g)) \in G \times \Sigma_n\}_{g \in G}$, the projection $\pi_G : G \times \Sigma_n \rightarrow G$ restricts to an isomorphism $\Gamma_\alpha \xrightarrow{\sim} G$. The α -twisted action is induced by the pullback along this isomorphism.

1.4. Chaotic-Forgetful Adjunction. Given any small category $\mathcal{A} \in \mathbf{Cat}$, its object $\text{Ob}(\mathcal{A})$ forms a set; moreover, any functor $F : \mathcal{A} \rightarrow \mathcal{B}$ also has an underlying map of objects $F : \text{Ob}(\mathcal{A}) \rightarrow \text{Ob}(\mathcal{B})$. Conversely, given a set, it has certain category structure:

Definition 1.10. Given a set X , define $\mathcal{E}X$ to be the *chaotic category* with underlying object set $\text{Ob}(\mathcal{E}X) := X$, by encoding a unique isomorphism $x \xrightarrow{\sim} y$ between any element $x, y \in X$. The composition

$$(1.11) \quad (x \xrightarrow{\sim} y \xrightarrow{\sim} z) = (x \xrightarrow{\sim} z)$$

using the uniqueness.

Given a set map $f : X \rightarrow Y$, define the functor $\mathcal{E}f : \mathcal{E}X \rightarrow \mathcal{E}Y$ so that the underlying object map is f , and for each morphism $x \xrightarrow{\sim} y$ in $\mathcal{E}X$, $\mathcal{E}f(x \rightarrow y) := f(x) \rightarrow f(y)$, the unique morphism between $f(x), f(y)$ in $\mathcal{E}Y$.

Inspecting the definition, we see $\mathcal{E} : \mathbf{Set} \rightarrow \mathbf{Cat}$ is a well-defined functor, called the *chaotic functor*.

Notice if a functor $F : \mathcal{A} \rightarrow \mathcal{B}$ has a chaotic target $\mathcal{B}$, then all the morphisms $f : X \rightarrow Y$ in $\mathcal{A}$ must be sent to the unique morphism $F(X) \rightarrow F(Y)$ in $\mathcal{B}$, hence F is dictated by the object map $F : \text{Ob}(\mathcal{A}) \rightarrow \text{Ob}(\mathcal{B})$. Such uniqueness specifies an adjunction $(\text{Ob} \dashv \mathcal{E})$.

The consequence is the following natural isomorphism of homset for any two sets X, Y :

$$(1.12) \quad \mathbf{Set}(X, Y) = \mathbf{Set}(\text{Ob}(\mathcal{E}X), Y) \cong \mathbf{Cat}(\mathcal{E}X, \mathcal{E}Y)$$

Remark 1.13. Let $\mathbf{Set}_G$ denotes the category of (left) G -sets, with all maps between G -sets (not necessarily equivariant) being the morphisms, the above adjunction can be promoted to $\text{Ob} : \mathbf{Cat}_G \rightarrow \mathbf{Set}_G$, and $\mathcal{E} : \mathbf{Set}_G \rightarrow \mathbf{Cat}_G$. Similar notions hold when transfer from left to right G -sets/categories.

In this exposition we only need $\mathcal{E}$ to be a right adjoint, there are richer details demonstrated in [3],

2. OPERADS, OPERADIC ALGEBRA, AND OPERAD $\mathcal{P}$

2.1. Symmetric Monoidal Categories.

Definition 2.1. A category $\mathcal{V}$ is a *symmetric monoidal category*, if there is a bifunctor $\otimes : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$ (called *monoidal product*), and an object $\kappa \in \mathcal{V}$ (called *unit/monoidal unit*), together with the following natural isomorphisms:

- The *associator* $a : (-_1 \otimes -_2) \otimes -_3 \xrightarrow{\sim} -_1 \otimes (-_2 \otimes -_3)$.
- The *left unit* $\lambda : \kappa \otimes - \xrightarrow{\sim} \text{Id}_{\mathcal{V}}$ and the *right unit* $\rho : - \otimes \kappa \xrightarrow{\sim} \text{Id}_{\mathcal{V}}$.
- The *Symmetric Braiding* $B : -_1 \otimes -_2 \xrightarrow{\sim} -_2 \otimes -_1$, with extra property $B_{Y,X} : Y \otimes X \rightarrow X \otimes Y$ equals to $B_{X,Y}^{-1} : X \otimes Y \rightarrow Y \otimes X$.

They also satisfy the following axioms:

- **(Triangle)**

$$\begin{array}{ccc} (X \otimes \kappa) \otimes Y & \xrightarrow{a_{X,\kappa,Y}} & X \otimes (\kappa \otimes Y) \\ \rho_X \otimes \text{id}_Y \searrow & & \swarrow \text{id}_X \otimes \lambda_Y \\ & X \otimes Y & \end{array}$$

- **(Pentagon)**

$$\begin{array}{ccc} ((X \otimes Y) \otimes Z) \otimes W & \xrightarrow{a_{X \otimes Y, Z, W}} & (X \otimes Y) \otimes (Z \otimes W) \\ \downarrow a_{X,Y,Z} \otimes \text{id}_W & & \downarrow a_{X,Y,Z \otimes W} \\ (X \otimes (Y \otimes Z)) \otimes W & & X \otimes (Y \otimes (Z \otimes W)) \\ \searrow a_{X,Y \otimes Z, W} & & \swarrow \text{id}_X \otimes a_{Y,Z,W} \\ & X \otimes ((Y \otimes Z) \otimes W) & \end{array}$$

- **(Hexagon)**

$$\begin{array}{ccccc}
(X \otimes Y) \otimes Z & \xrightarrow{a_{X,Y,Z}} & X \otimes (Y \otimes Z) & \xrightarrow{B_{X,Y \otimes Z}} & (Y \otimes Z) \otimes X \\
\downarrow B_{X,Y} \otimes \text{Id}_Z & & & & \downarrow a_{Y,Z,X} \\
(Y \otimes X) \otimes Z & \xrightarrow{a_{Y,X,Z}} & Y \otimes (X \otimes Z) & \xrightarrow{\text{Id}_Y \otimes B_{X,Z}} & Y \otimes (Z \otimes X)
\end{array}$$

Based on *MacLane's Coherence Theorem*, the above two axioms guarantees all finite sequences of a, λ, ρ and their inverses agree, namely there is a unique isomorphism $P_1 \rightarrow P_2$, where P_i are (possibly) different orders of performing product on $X_1, \dots, X_n \in \mathcal{V}$. For details, cf. [11].

The category of sets **Set**, small categories **Cat** in 1.3, and small G -categories **Cat_G** in 1.8 are central examples for this, where the cartesian product $\mathcal{A} \times \mathcal{B}$ serves as monoidal product, and the terminal set/category $*$ serves as units for both.

2.2. Operads, Operadic Algebras.

Definition 2.2. Given symmetric monoidal category $\mathcal{V}$, a Σ *operad* $\mathcal{C}$ in $\mathcal{V}$ is a collection of objects $\{\mathcal{C}(j) \in \mathcal{V}\}_{j \in \mathbb{N}}$, each $\mathcal{C}(j)$ with a right Σ_j -action (a group homomorphism $\Sigma_j^{\text{op}} \rightarrow \text{Aut}_{\mathcal{V}}(\mathcal{C}(j))$), a *unit map* $\eta : 1 \rightarrow \mathcal{C}(1)$, together with composition maps

$$\gamma : \mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes \dots \otimes \mathcal{C}(j_k) \rightarrow \mathcal{C}(j_1 + \dots + j_k)$$

that satisfies the following axioms:

- *Unit:*

$$\begin{array}{ccc}
\mathcal{C}(j) \otimes \kappa^{\otimes j} & \xrightarrow{\sim} & \mathcal{C}(j) \\
\searrow \text{id}_{\mathcal{C}(j)} \otimes \eta^{\otimes j} & & \nearrow \gamma \\
& \mathcal{C}(j) \otimes \mathcal{C}(1)^{\otimes j} &
\end{array}
\quad
\begin{array}{ccc}
\kappa \otimes \mathcal{C}(j) & \xrightarrow{\sim} & \mathcal{C}(j) \\
\searrow \eta \otimes \text{id}_{\mathcal{C}(j)} & & \nearrow \gamma \\
& \mathcal{C}(1) \otimes \mathcal{C}(j) &
\end{array}$$

- *Associativity:* Fix natural numbers $k, j = \sum_{s=1}^k j_s$, define index $g_s := j_1 + \dots + j_s$ for $1 \leq s \leq k$. For each $i_1, \dots, i_{g_k} = i_j$, with sum $i = \sum_{r=1}^j i_r$, and $h_s := i_{g_{(s-1)}+1} + \dots + i_{g_s}$ (so $i = \sum_{s=1}^k h_s$ also), the following diagram commutes:

$$\begin{array}{ccc}
\left[\mathcal{C}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{C}(j_s) \right) \right] \otimes \left(\bigotimes_{r=1}^j \mathcal{C}(i_r) \right) & \xrightarrow{\gamma \otimes \text{id}} & \mathcal{C}(j) \otimes \left(\bigotimes_{r=1}^j \mathcal{C}(i_r) \right) \\
\downarrow \text{shuffle} & & \downarrow \gamma \\
& & \mathcal{C}(i) \\
& & \uparrow \gamma \\
\mathcal{C}(k) \otimes \left[\bigotimes_{s=1}^k \left(\mathcal{C}(j_s) \otimes \left(\bigotimes_{q=1}^{j_s} \mathcal{C}(i_{g_{(s-1)}+q}) \right) \right) \right] & \xrightarrow{\text{id}_{\mathcal{C}(k)} \otimes (\bigotimes_s \gamma)} & \mathcal{C}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{C}(h_s) \right)
\end{array}$$

- *Equivariance:*

- Given $\sigma \in \Sigma_k$, $j_1 + \dots + j_k = \Sigma_j$, let $\sigma(j_1, \dots, j_k) \in \Sigma_j$ denotes “permutation of k -blocks of letters”, by $(j_1, \dots, j_k) \mapsto (j_{\sigma^{-1}(1)}, \dots, j_{\sigma^{-1}(k)})$. Then, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r) \right) & \xrightarrow{\sigma \otimes \sigma^{-1}} & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_{\sigma(r)}) \right) \\ \gamma \downarrow & & \downarrow \gamma \\ \mathcal{C}(j) & \xrightarrow{\sigma(j_{\sigma(1)}, \dots, j_{\sigma(k)})} & \mathcal{C}(j) \end{array}$$

- Given $\tau_r \in \Sigma_{j_r}$, denote $\tau_1 \oplus \dots \oplus \tau_k \in \Sigma_j$ by letting each τ_r permuting the j_r th block in j . Then, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r) \right) & \xrightarrow{\text{id}_{\mathcal{C}(k)} \otimes (\bigotimes_r \tau_r)} & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r) \right) \\ \gamma \downarrow & & \downarrow \gamma \\ \mathcal{C}(j) & \xrightarrow{\tau_1 \oplus \dots \oplus \tau_k} & \mathcal{C}(j) \end{array}$$

If $\mathcal{C}(0) = \kappa$ (with $\gamma : \mathcal{C}(0) \rightarrow \mathcal{C}$ being $\text{id}_{\mathcal{C}(0)}$), the operad is *reduced*.

Definition 2.3. Given an operad $\mathcal{C}$ over a symmetric monoidal category $\mathcal{V}$, a $\mathcal{C}$ -*algebra* is an object $A \in \mathcal{V}$, with collections of morphisms $\theta : \mathcal{C}(j) \otimes A^j \rightarrow A$, that satisfies the following axioms:

- *Associativity*: Given $j = \sum_{i=1}^k j_i$, the following diagram commutes:

$$\begin{array}{ccc} \left[\mathcal{C}(k) \otimes \left(\bigotimes_{i=1}^k \mathcal{C}(j_i) \right) \right] \otimes A^{\otimes j} & \xrightarrow{\gamma \otimes \text{id}_{A^{\otimes j}}} & \mathcal{C}(j) \otimes A^{\otimes j} \\ \downarrow \text{shuffle} & & \downarrow \theta \\ \mathcal{C}(k) \otimes \left[\bigotimes_{i=1}^k (\mathcal{C}(j_i) \otimes A^{\otimes j_i}) \right] & \xrightarrow{\text{id}_{\mathcal{C}(k)} \otimes \theta^k} & \mathcal{C}(k) \otimes A^{\otimes k} \\ & & \uparrow \theta \\ & & A \end{array}$$

(Note: This part uses the isomorphism $A^{\otimes j} \cong \bigotimes_{i=1}^k A^{\otimes j_i}$).

- *Unit*: The following diagram commutes:

$$\begin{array}{ccc} \kappa \otimes A & \xrightarrow{\sim} & A \\ & \searrow \eta \otimes \text{id}_A & \nearrow \theta \\ & \mathcal{C}(1) \otimes A & \end{array}$$

- *Equivariance*: Given any permutation $\sigma \in \Sigma_j$, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{C}(j) \otimes A^{\otimes j} & \xrightarrow{\sigma \otimes \sigma^{-1}} & \mathcal{C}(j) \otimes A^{\otimes j} \\ & \searrow \theta & \swarrow \theta \\ & A & \end{array}$$

Example 2.4. The *associativity operad* $\mathcal{M}$ in **Set** is defined as $\mathcal{M}(j) := \Sigma_j$ (the symmetric group for cardinality j), which endows a natural right Σ_j -action via right multiplication. The operadic structure map is given by

(2.5)

$$\gamma : \Sigma_k \times \prod_{r=1}^k \Sigma_{j_r} \rightarrow \Sigma_j, \quad \gamma(\sigma; \tau_1, \dots, \tau_k) := \tau_{\sigma^{-1}(1)} \oplus \dots \oplus \tau_{\sigma^{-1}(k)} \circ \sigma(j_{\sigma(1)}, \dots, j_{\sigma(k)})$$

where $j = \sum_r j_r$, $\sigma(j_1, \dots, j_k) \in \Sigma_j$ denotes the block permutation of $j_1, \dots, j_k$ within j (in order), and $\tau_1 \oplus \dots \tau_k$ is the image of the embedding $\oplus : \prod_{r=1}^k \Sigma_{j_r} \rightarrow \Sigma_j$, by viewing the r th block as finite set of cardinality j_r . Inspecting the definitions, the algebra over $\mathcal{M}$ precisely describes the set theoretic monoids. For more details, cf. [3] Section 4.1.

2.3. Permutativity Operad $\mathcal{P}$. Now, recall that the chaotic functor $\mathcal{E} : \mathbf{Set} \rightarrow \mathbf{Cat}$ is a right adjoint of $\text{Ob} : \mathbf{Cat} \rightarrow \mathbf{Set}$, in particular it preserves products. Hence, the chaotic categorification $\mathcal{P}(j) := \mathcal{E}\mathcal{M}(j)$ also has a natural operadic structure, via the identification

$$(2.6) \quad \mathcal{E}\mathcal{M}(k) \times \prod_{r=1}^k \mathcal{E}\mathcal{M}(j_r) \cong \mathcal{E} \left(\mathcal{M}(k) \times \prod_{r=1}^k \mathcal{M}(j_r) \right) \xrightarrow{\mathcal{E}\gamma} \mathcal{E}\mathcal{M}(j).$$

This is called the *permutativity operad in $\mathbf{Cat}$* , or the *Barratt-Eccles operad*. As an abuse of notation, the operadic structure map of $\mathcal{P}$ is also denoted γ .

3. G -OPERAD $\mathcal{P}_G$ AND ITS PSEUDOALGEBRA

3.1. The G -Operad $\mathcal{P}_G$. If view each $\mathcal{P}(j)$ as a G -trivial G -category, it's also an operad in $\mathbf{Cat}_G$. Its underlying algebra is called the *naive permutativity G -category*, as specified in [3] Section 4.1, it produces a naive G -spectrum under equivariant infinite loop space theory. In the same source Section 4.2, a richer equivariant structure is introduced which produces the *equivariant Barratt Eccles G -operad* $\mathcal{P}_G$ as follow:

Definition 3.1. The j th category $\mathcal{P}_G(j) := \mathbf{Cat}(\mathcal{E}G, \mathcal{P}(j))$, with the operadic structure map determined by the fact that $\mathbf{Cat}(\mathcal{E}G, -)$ preserves product:

(3.2)

$$\mathbf{Cat}(\mathcal{E}G, \mathcal{P}(k)) \times \prod_{r=1}^k \mathbf{Cat}(\mathcal{E}G, \mathcal{P}(j_r)) \cong \mathbf{Cat} \left(\mathcal{E}G, \mathcal{P}(k) \times \prod_{r=1}^k \mathcal{P}(j_r) \right) \xrightarrow{\gamma \circ -} \mathbf{Cat}(\mathcal{E}, \mathcal{P}(j))$$

Abuse of notation, we'll denote the operadic structure map as γ again.

Another formulation can be done by defining $P_G(j) := \mathbf{Set}(G, \Sigma_j)$ in **Set**, then utilize chaotic functor promoting this to a chaotic operad $\mathcal{E}\mathbf{Set}(G, \Sigma) \cong \mathcal{P}_G$.

3.2. The 2-Category $\mathcal{P}_G\text{-PsAlg}$. The general notion of operadic pseudoalgebra is defined in [2], [5], and [8]. Here we only take the case when the operad is $\mathcal{P}_G$, together with some description of G -equivariance.

Definition 3.3 ($\mathcal{P}_G$ -pseudoalgebra). A $\mathcal{P}_G$ -pseudoalgebra is a G -category $\mathcal{A}$ and operad action functors

$$\theta_j : \mathcal{P}_G(j) \times \mathcal{A}^j \rightarrow \mathcal{A}$$

for each $n \in \mathbb{N}$, together with an invertible natural transformation ϕ_1 in the diagram

$$(3.4) \quad \begin{array}{ccc} \mathcal{A} & \xrightarrow{\iota} & \mathcal{P}_G(1) \times \mathcal{A} \\ & \searrow \phi_1 & \downarrow \theta_1 \\ & \text{Id}_{\mathcal{A}} & C \end{array}$$

An invertible natural transformation $\phi = \phi(k; j_1, \dots, j_k)$ for each diagram

$$(3.5) \quad \begin{array}{ccc} \mathcal{P}_G(k) \times \left(\prod_{r=1}^k \mathcal{P}_G(j_r) \times \mathcal{A}^{j_r} \right) & \xrightarrow{\text{id} \times \prod_r \theta_{j_r}} & \mathcal{P}_G(k) \times \mathcal{A}^k \\ \downarrow \text{shuffle} & \searrow \phi & \downarrow \theta_k \\ \mathcal{P}_G(k) \times \prod_{r=1}^k \mathcal{P}_G(j_r) \times \mathcal{A}^{j_r} & \xrightarrow{\gamma \times \text{id}} & \mathcal{P}_G(j_+) \times \mathcal{A}^{j_+} \end{array}$$

such that the following strict equivariance hold for θ_j and ϕ :

- (i) **(Equivariance of θ_j)** The following two diagrams commutes for $\sigma \in \Sigma_j$ and $g \in G$:

$$(3.6) \quad \begin{array}{ccc} \mathcal{P}_G(j) \times \mathcal{A}^j & \xrightarrow{\sigma \times \sigma^{-1}} & \mathcal{P}_G(j) \times \mathcal{A}^j \\ & \searrow \theta_j & \swarrow \theta_j \\ & \mathcal{A} & \end{array} \quad , \quad \begin{array}{ccc} \mathcal{P}_G(j) \times \mathcal{A}^j & \xrightarrow{g \times g^{-1}} & \mathcal{P}_G(j) \times \mathcal{A}^j \\ \theta_j \downarrow & & \downarrow \theta_j \\ \mathcal{A} & \xrightarrow{g^{-1}} & \mathcal{A} \end{array}$$

The left is the operadic equivariance, the right is G -equivariance by asserting a product G -action on $\mathcal{P}(j) \times \mathcal{A}^j$.

- (ii) **(Equivariance of ϕ)** The following equalities hold when given $\sigma \in \Sigma_k$, and $\tau_r \in \Sigma_{j_r}$:

$$\phi(k; j_1, \dots, j_k) = \phi(k; j_{\sigma(1)}, \dots, j_{\sigma(k)}) \circ (\sigma \times \sigma^{-1})$$

$$\phi(k; j_1, \dots, j_k) = \phi(k; j_1, \dots, j_k) \circ (\text{id} \times \prod_r (\tau_r \times \tau_r^{-1}))$$

They also need to satisfy the following coherence properties:

- (iii) **(Operadic Composition)** The following pasting diagrams are equal, given $j_+ = \sum j_r$, $i_r = \sum_s i_{r,s}$, and $i_+ = \sum_{r,s} i_{r,s}$ with $1 \leq r \leq k$ and $1 \leq s \leq j_r$:

$$\begin{array}{ccccc}
 \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \prod_s (\mathcal{P}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}})) & \xrightarrow{\text{id} \times \prod (\text{id} \times \theta_{i_{r,s}})} & \mathcal{Q}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \mathcal{A}^{j_r}) & & \\
 \downarrow \gamma \times \text{id} & & \downarrow \gamma & \searrow \text{id} \times \prod \theta_{j_r} & \\
 \mathcal{P}_G(j_+) \times \prod_{r,s} (\mathcal{P}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}}) & \xrightarrow{\text{id} \times \prod \theta_{i_{r,s}}} & \mathcal{P}_G(j_+) \times \mathcal{A}^{j_+} & \xrightarrow{\phi} & \mathcal{P}_G(k) \times \mathcal{A}^k \\
 & \searrow \gamma \times \text{id} & \downarrow \phi & \searrow \theta_{j_+} & \downarrow \theta_k \\
 & & \mathcal{P}_G(i_+) \times \mathcal{A}^{i_+} & \xrightarrow{\theta_{j_+}} & \mathcal{A}
 \end{array}$$

equals

$$\begin{array}{ccccc}
 \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \prod_s (\mathcal{P}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}})) & \xrightarrow{\text{id} \times \prod (\text{id} \times \prod \theta_{i_{r,s}})} & \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \mathcal{A}^{j_r}) & & \\
 \downarrow \gamma \times \text{id} & \searrow \text{id} \times \prod (\gamma \times \text{id}) & \downarrow \text{id} \times \prod \phi & \searrow \text{id} \times \prod \theta_{j_r} & \\
 \mathcal{P}_G(j_+) \times \prod_{r,s} (\mathcal{P}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}}) & & \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(i_r) \times \mathcal{A}^{i_r}) & \xrightarrow{\text{id} \times \prod \gamma} & \mathcal{P}_G(k) \times \mathcal{A}^k \\
 & \searrow \gamma \times \text{id} & \downarrow \gamma \times \text{id} & \searrow \phi & \downarrow \theta_k \\
 & & \mathcal{P}_G(i_+) \times \mathcal{A}^{i_+} & \xrightarrow{\theta_{i_+}} & \mathcal{A}
 \end{array}$$

The commuting square above is due to associativity of operad.

- (iv) **(Operadic Identity)** The following two pairs of pasting diagrams are equal:

$$\begin{array}{ccc}
 \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id} \times \iota^k} & \mathcal{P}_G(k) \times (\mathcal{P}_G(1) \times \mathcal{A})^k \xrightarrow{\text{id} \times \theta_1^k} \mathcal{P}_G(k) \times \mathcal{A}^k \\
 & \downarrow \text{shuffle} & \downarrow \theta_k \\
 & \mathcal{P}_G(k) \times \mathcal{P}_G(1)^k \times \mathcal{A}^k & \downarrow \gamma \times \text{id} \\
 & \downarrow \gamma \times \text{id} & \swarrow \phi \\
 \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} & \mathcal{A}
 \end{array}$$

equals

$$\begin{array}{ccccc}
 & & \mathcal{P}_G(k) \times (\mathcal{P}_G(1) \times \mathcal{A})^k & & \\
 & \nearrow \text{id} \times \iota^k & \downarrow \text{id} \times \prod \phi & \nwarrow \text{id} \times \theta_1^k & \\
 \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\quad \text{id} \quad} & \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} & \mathcal{A} \\
 \downarrow \text{id} \times \iota^k & \searrow \iota^k & \nearrow \gamma \times \text{id} & & \\
 \mathcal{P}_G(k) \times (\mathcal{P}_G(1) \times \mathcal{A})^k & \xrightarrow{\text{shuffle}} & \mathcal{P}_G(k) \times \mathcal{P}_G(1)^k \times \mathcal{A}^k & &
 \end{array}$$

(The two triangles right above commutes due to the unit axioms of operad).

Also:

$$\begin{array}{ccccc}
 \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\iota} & \mathcal{P}_G(1) \times \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id} \times \theta_k} & \mathcal{P}_G(1) \times \mathcal{A} \\
 & \searrow \text{id} & \downarrow \gamma \times \text{id} & \swarrow \phi & \downarrow \theta_1 \\
 & & \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} & \mathcal{A}
 \end{array}$$

equals

$$\begin{array}{ccc}
 \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\iota} & \mathcal{P}_G(1) \times \mathcal{P}_G(k) \times \mathcal{A}^k \\
 \downarrow \theta_k & & \downarrow \text{id} \times \theta_k \\
 \mathcal{A} & \xrightarrow{\iota} & \mathcal{P}_G(1) \times \mathcal{A} \\
 & \swarrow \phi_1 & \downarrow \theta_1 \\
 & & \mathcal{A}
 \end{array}$$

The first diagram's triangle commutes due to the operadic unit axiom.

Remark 3.7. In [2] Corner and Gurski includes unital natural isomorphisms, while in [5] it requires strict unitality. The main result in the nonequivariant case of [8] alleviates the result from strict unitality.

Definition 3.8 ($\mathcal{P}_G$ -pseudomorphism). A $\mathcal{P}_G$ -pseudomorphism $(\mathbb{F}, \zeta) : (\mathcal{A}, \theta, \phi) \rightarrow (\mathcal{B}, \vartheta, \varphi)$ consists of functor $\mathbb{F} : \mathcal{A} \rightarrow \mathcal{B}$, such that for each j in the following diagram commutes up to an invertible natural transformation ζ_j :

$$\begin{array}{ccc}
 \mathcal{P}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{P}_G(j) \times \mathcal{B}^j \\
 \downarrow \theta_j & \swarrow \zeta_j & \downarrow \vartheta_j \\
 \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
 \end{array}$$

Moreover, they satisfy the following properties:

- (i) **(Equivariance of ζ)** For each $j, \zeta_j = \zeta_j \circ (\sigma \times \sigma^{-1})$ for all $\sigma \in \Sigma_j$. **Check G -equivariance**

(ii) **(Preserves Identity)** The following two pasting diagrams are equal:

$$\begin{array}{ccc}
 \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B} \\
 \downarrow \iota & & \downarrow \iota \\
 \mathcal{P}_G(1) \times \mathcal{A} & \xrightarrow{\text{id} \times \mathbb{F}} & \mathcal{P}_G(1) \times \mathcal{B} \\
 \searrow \theta_1 & \nearrow \zeta_1^{-1} & \searrow \varphi_1 \\
 & \mathcal{A} & \xrightarrow{\mathbb{F}} \mathcal{B}
 \end{array}
 \quad = \quad
 \begin{array}{ccc}
 \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B} \\
 \downarrow \iota & \searrow \text{id} & \downarrow \text{id} \\
 \mathcal{P}_G(1) \times \mathcal{A} & \xrightarrow{\theta_1} & \mathcal{A} \xrightarrow{\mathbb{F}} \mathcal{B}
 \end{array}$$

(iii) **(Preserves Composition)** The following two pasting diagrams are equal:

$$\begin{array}{ccccc}
 \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod (\text{id} \times \mathbb{F}^{j_r})} & \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \mathcal{B}^{j_r}) & \xrightarrow{\text{id} \times \prod \vartheta_{j_r}} & \mathcal{P}_G(k) \times \mathcal{B}^k \\
 \downarrow \gamma & \searrow \text{id} \times \prod \theta_{j_r} & \swarrow \text{id} \times \prod \zeta_{j_r} & \searrow \text{id} \times \mathbb{F}^k & \downarrow \vartheta_k \\
 & \mathcal{P}_G(k) \times \mathcal{A}^k & & & \mathcal{P}_G(k) \times \mathcal{B}^k \\
 & \swarrow \phi & \downarrow \theta_k & \nearrow \zeta_k & \downarrow \vartheta_k \\
 \mathcal{P}_G(j) \times \mathcal{A}^j & & \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B} \\
 & \searrow \theta_j & & & \\
 & & & &
 \end{array}$$

equals

$$\begin{array}{ccccc}
 \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod (\text{id} \times \mathbb{F}^{j_r})} & \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \mathcal{B}^{j_r}) & \xrightarrow{\text{id} \times \prod \vartheta_{j_r}} & \mathcal{P}_G(k) \times \mathcal{B}^k \\
 \downarrow \gamma & & \downarrow \gamma & \searrow \text{id} \times \prod \theta_{j_r} & \downarrow \vartheta_k \\
 \mathcal{P}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{P}_G(j) \times \mathcal{B}^j & \swarrow \phi & \downarrow \vartheta_k \\
 & \searrow \theta_j & \swarrow \zeta_j & \searrow \vartheta_j & \\
 & & \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
 \end{array}$$

Definition 3.9 ($\mathcal{P}_G$ -2-cells). Let $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : (\mathcal{A}, \theta, \phi) \rightarrow (\mathcal{B}, \vartheta, \varphi)$ be two $\mathcal{P}_G$ -pseudomorphisms, a $\mathcal{P}_G$ -2-cell $\lambda : (\mathbb{F}, \zeta) \Rightarrow (\mathbb{G}, \xi)$ is a natural transformation

$\lambda : \mathbb{F} \rightarrow \mathbb{G}$, such that the following two pasting diagrams are equal:

$$\begin{array}{ccc}
 \mathcal{P}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{P}_G(j) \times \mathcal{B}^j \\
 \downarrow \theta_j & \searrow \text{id} \times \lambda^j & \downarrow \vartheta_j \\
 \mathcal{A} & \xrightarrow{\mathbb{G}} & \mathcal{B}
 \end{array}
 \quad = \quad
 \begin{array}{ccc}
 \mathcal{P}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{P}_G(j) \times \mathcal{B}^j \\
 \downarrow \theta_j & \searrow \zeta_j & \downarrow \vartheta_j \\
 \mathcal{A} & \xrightarrow{\lambda} & \mathcal{B}
 \end{array}$$

$\mathbb{F}$ (curved arrow from $\mathcal{A}$ to $\mathcal{B}$ in the right diagram)
 $\mathbb{G}$ (straight arrow from $\mathcal{A}$ to $\mathcal{B}$ in the right diagram)

4. CATEGORY $\mathcal{F}_G$ AND ITS PSEUDOALGEBRAS

Within the equivariant infinite loop space machine, the purpose for $\mathcal{F}_G$ -pseudoalgebras (after strictifying to algebras) are to construct $\mathcal{F}_G$ -spaces, which are collections of G -spaces indexed over finite based G -sets (and morphisms between them encoding extra structure). The $\mathcal{F}_G$ -spaces are primary inputs in producing a genuine G -spectra, and the other main character we wish to study the correspondance. For details of $\mathcal{F}_G$ -spaces, cf. [14] Section 1 (which is named after Γ_G instead of $\mathcal{F}_G$).

The following definitions are equivariant generalizations of $\mathcal{F}$ -pseudofunctors in [8], and we'll utilize similar notations.

Definition 4.1. $\mathcal{F}_G$ denotes the *category of based finite sets*, where the objects are $\mathbf{n}^\alpha$, set wise being the finite set $\{0, 1, \dots, n\}$ with 0 fixed as the basepoint, and $\alpha : G \rightarrow \Sigma_n$ is a based G -action on the set.

The homset $\mathcal{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ consists of all the based set map, which the homset also endows with a G -action via conjugation. From now on, we'll view $\mathcal{F}_G$ as a 2-category, where each hom-category $\mathcal{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ is discrete with only identity 2-cells.

Furthermore, we define the following:

- $\Pi_G \subset \mathcal{F}_G$ denotes the subcategory where each morphism $\phi : \mathbf{n}^\alpha \rightarrow \mathbf{m}^\beta$ satisfies $|\phi^{-1}(j)| = 0, 1$ for all $j \neq 0$.
- $\Lambda_G, \Lambda_G^\perp, \Sigma_G \subset \Pi_G$ denotes the subcategory where each morphism is injective, surjective, and bijective respectively (in particular, $\Lambda_G \cap \Lambda_G^\perp = \Sigma_G$).
- $\mathcal{E}_G \subset \mathcal{F}_G$ denotes the subcategory where each morphism satisfies $\phi^{-1}(0) = \{0\}$.

We also denote the map $\phi_{\mathbf{n}^\alpha} : \mathbf{n}^\alpha \rightarrow \mathbf{1}$ (with $j \mapsto 1$ for all $j \neq 0$) in $\mathcal{E}$ as the *canonical product*.

4.1. 2-Category $\mathcal{F}_G$ -PsAlg.

Definition 4.2 ($\mathcal{F}_G$ -pseudoalgebra). An $\mathcal{F}_G$ -pseudoalgebra, is a pseudofunctor $\mathcal{B} : \mathcal{F}_G \rightarrow \mathbf{Cat}_G$ (cf. Definition 1.4), such that restricting to the subcategory $\Pi_G \subset \mathcal{F}_G$, $\mathcal{B}$ is a strict functor. Moreover, we require it to be reduced, namely $\mathcal{B}(0) = *$ (the terminal G -category), and each hom-functor

$$\mathcal{B}_{\mathbf{n}^\alpha, \mathbf{m}^\beta} : \mathcal{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta) \rightarrow \mathbf{Cat}_G(\mathcal{B}(\mathbf{n}^\alpha), \mathcal{B}(\mathbf{m}^\beta))$$

is a G -functor.

Let $\delta_i : \mathbf{n}^\alpha \rightarrow \mathbf{1}$ be the i th projection (namely $i \mapsto 1$, and any $i \neq j \mapsto 0$), since it belongs to Π_G , it generates a functor $\mathcal{B}\delta_i : \mathcal{B}(\mathbf{n}^\alpha) \rightarrow \mathcal{B}(\mathbf{1})$. Hence, by universality of product, the collection $\mathcal{B}\delta_i$ ($1 \leq i \leq n$) induces a functor $\delta : \mathcal{B}(\mathbf{n}^\alpha) \rightarrow \mathcal{B}(\mathbf{1})^{\mathbf{n}^\alpha}$ (where $\mathcal{B}(\mathbf{1})^{\mathbf{n}} \cong \mathcal{B}(\mathbf{1})^{\mathbf{n}^\alpha}$ nonequivariantly).

Definition 4.3 ($\mathcal{F}_G$ -vs and $\mathcal{F}_{G,\mathbb{R}}$ -pseudoalgebra). An $\mathcal{F}_G$ -pseudoalgebra $\mathcal{B}$ is *very special* (or an $\mathcal{F}_{G,\text{vs}}$ -pseudoalgebra), if $\delta : \mathcal{B}(\mathbf{n}^\alpha) \rightarrow \mathcal{B}(\mathbf{1})^{\mathbf{n}^\alpha}$ is an isomorphism for all $\mathbf{n}^\alpha \in \mathcal{F}_G$, and $\mathcal{B}$ is *strictly special* (or an $\mathcal{F}_{G,\mathbb{R}}$ -pseudoalgebra), if all δ is in fact identity.

Definition 4.4 ($\mathcal{F}_G$ -1-cell). Given $\mathcal{A}, \mathcal{B}$ two $\mathcal{F}_G$ -pseudoalgebras, a morphism (or $\mathcal{F}_G$ -1-cell) between them is a pseudotransform $(\mathbb{F}, \zeta) : \mathcal{A} \rightarrow \mathcal{B}$ (cf. Definition 1.5), such that restricting to Π_G , it becomes an actual natural transformation. Moreover, it respects G -equivariance for every $g \in G$ and morphism $f \in \text{Mor}(\mathcal{F}_G)$:

$$\zeta(g \circ f \circ g^{-1}) = g \circ \zeta(f) \circ g^{-1}$$

Definition 4.5 ($\mathcal{F}_G$ -2-cell). Given $\tilde{\mathcal{A}}, \tilde{\mathcal{B}}$ two $\mathcal{F}_G$ -pseudoalgebras, and two $\mathcal{F}_G$ -1-cells $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : \tilde{\mathcal{A}} \rightarrow \tilde{\mathcal{B}}$, an $\mathcal{F}_G$ -2-cell is a modification $\lambda : \mathbb{F} \Rightarrow \mathbb{G}$ (cf. Definition 1.6), such that for any morphism $f : \mathbf{n}^\alpha \rightarrow \mathbf{m}^\beta$ in $\mathcal{F}_G$, the following two pasting diagrams are equal:

$$\begin{array}{ccc} \mathcal{A}(\mathbf{n}^\alpha) & \xrightleftharpoons[\mathbb{G}_{\mathbf{n}^\alpha}]{\lambda} & \mathcal{B}(\mathbf{n}^\alpha) \\ \mathcal{A}f \downarrow & \nearrow \xi f & \downarrow \mathcal{B}f \\ \mathcal{A}(\mathbf{m}^\beta) & \xrightarrow{\mathbb{G}_{\mathbf{m}^\beta}} & \mathcal{B}(\mathbf{m}^\beta) \end{array} = \begin{array}{ccc} \mathcal{A}(\mathbf{n}^\alpha) & \xrightarrow{\mathbb{F}_{\mathbf{n}^\alpha}} & \mathcal{B}(\mathbf{n}^\alpha) \\ \mathcal{B}f \downarrow & \nearrow \zeta f & \downarrow \mathcal{B}f \\ \mathcal{A}(\mathbf{m}^\beta) & \xrightleftharpoons[\mathbb{G}_{\mathbf{m}^\beta}]{\lambda} & \mathcal{B}(\mathbf{m}^\beta) \end{array}$$

5. OBSTACLES IN EQUIVARIANT CASE (ND)

The eventual goal is constructing an equivalence between $\mathcal{F}_{G,\mathbb{R}}$ -pseudoalgebra and $\mathcal{P}_G$ -pseudoalgebra (though it's still an open question). [8] provides a general construction for the non-equivariant case (or $G = \mathbf{1}$), here we discuss some potential steps and problems when generalizing the procedure to nonequivariant case.

5.1. Functor $\mathcal{P}_G\text{-PsAlg} \rightarrow \mathcal{F}_{G,\mathbb{R}}\text{-PsAlg}$. Given a $\mathcal{P}$ -pseudoalgebra $\mathcal{A}$, define object map $\mathcal{B} : \mathcal{F} \rightarrow \mathbf{Cat}$ by $\mathcal{B}(\mathbf{n}) := \mathcal{A}^{\mathbf{n}}$. For extending to a pseudofunctor (and restricting to functor on Π), the central concepts include:

- (i) Substituting the vanishing coordinates of maps $\psi : \mathbf{n} \rightarrow \mathbf{m}$ in Π using the specified basepoint $\theta_0 : * \cong \mathcal{P}(0) \rightarrow \mathcal{A}$. In particular, since each $j \in \mathbf{m}$ has preimage being empty or a singleton, define $\psi : \mathcal{A}^{\mathbf{n}} \rightarrow \mathcal{A}^{\mathbf{m}}$ by

$$\psi(a_1, \dots, a_n) := (a_{\psi^{-1}(1)}, \dots, a_{\psi^{-1}(m)}),$$

with $a_{\psi^{-1}(j)} := *$ (the basepoint) if $\psi^{-1}(j) = \emptyset$. This defines $\mathcal{B}$ to be a functor with source Π .

- (ii) Treating each canonical n -fold product $\phi_n : \mathbf{n} \rightarrow \mathbf{1}$ in $\mathcal{E}$ (with $\phi_n(j) = 1$ for all $j \neq 0$) using the n -fold product $e \times \mathcal{A} \hookrightarrow \mathcal{P}(n) \times \mathcal{A} \xrightarrow{\gamma} \mathcal{A}$.
- (iii) Finally, show the decomposition laws, that maps in $\mathcal{E}$ are compositions of permutation and wedge of canonical products (when fixing a specific order, it's unique), while maps in $\mathcal{F}$ decomposes into a projection and a map in $\mathcal{E}$. Using the laws to construct functors $\mathcal{A}^n \rightarrow \mathcal{A}^m$ using each map ψ .

The coherent natural isomorphisms for $\mathcal{P}$ -pseudoalgebra provides the pseudo-functoriality modulo extensive diagram chasing.

Step (i) generalizes to the equivariant case, as each $\mathcal{P}_G$ -pseudoalgebra $\mathcal{A}$ has a natural basepoint specified by $\theta_0 : * \cong \mathcal{P}_G(0) \rightarrow \mathcal{A}$, each $\mathcal{A}^{\mathbf{n}^\alpha} \cong \mathcal{A}^n$ nonequivariantly, and the maps in Π_G includes all nonequivariant ones, hence the same construction carries over.

For (ii), each canonical product $\phi_{\mathbf{n}^\alpha} : \mathbf{n}^\alpha \rightarrow \mathbf{1}$ has a choice given by $\alpha \times \mathcal{A}^n \hookrightarrow \mathcal{P}_G(n) \times \mathcal{A}^n \xrightarrow{\theta_n} \mathcal{A}$ by treating α as an object in $\mathcal{P}_G(n) = \mathbf{Cat}(\mathcal{E}G, \mathcal{E}\Sigma_n) \cong \mathbf{Set}(G, \Sigma_n)$.

For (iii), the main problem in equivariant case is: Not every map can be expressed as wedges of canonical product, as the preimage of an element may not necessarily be wedges of G -orbits. So, potentially some choices of maps needed to be made, and more combinatorial details are awaiting.

5.2. Functor $\mathcal{P}_G\text{-PsAlg} \leftarrow \mathcal{F}_{G,\mathbb{R}}\text{-PsAlg}$. For the converse, given an $\mathcal{F}_{\mathbb{R}}$ -pseudo-functor $\mathcal{B}$, define the category $\mathcal{A} := \mathcal{B}(\mathbf{1})$, then each canonical product $\phi_n : \mathbf{n} \rightarrow \mathbf{1}$ (by $j \mapsto 1$ for all $j \neq 0$) generates an n -fold product $\phi_n : \mathcal{A}^n \rightarrow \mathcal{A}$. By appealing to more categorial details, this defines $\mathcal{A}$ as a $\mathcal{P}$ -pseudoalgebra.

The main obstacle in the equivariant case is the non-finiteness of G -operad $\mathcal{P}_G$ when $G \neq \mathbf{1}$: In [1]

5.3. Other Potential Direction.

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